GEOG 482 - Lab 4: Part 2
Tara Disser

Eugene, Oregon E-Scooter Hub Location Identification Report

Introduction

In order to identify suitable locations for e-scooter hubs in Eugene, we must first determine where the highest levels of demand exist. In order to do this, I used population density, residency, business, and workplace points, and bikeway and crosswalk data to determine where it makes most sense for e-scooter hubs to be placed. These input data layers were split into several models which output classified raster layers with values indicating levels of demand/suitability for an e-scooter hub for each cell, and finally combined to create a map of the study area depicting areas where e-scooter hub placement would be most suitable.

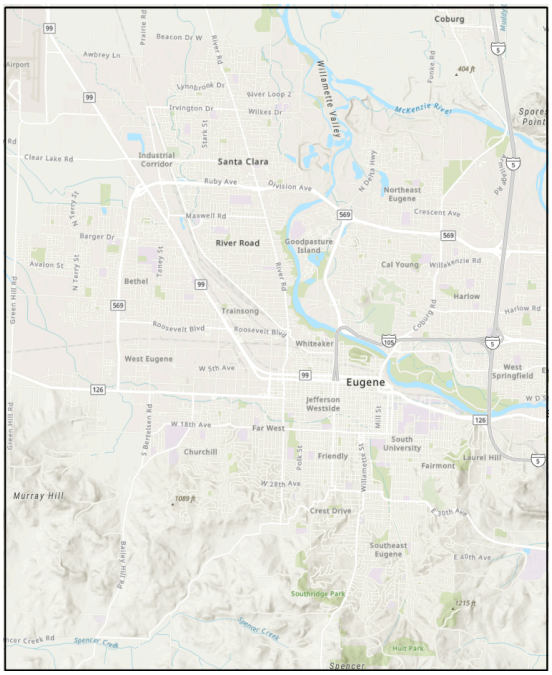


Figure 1: Study Area (Eugene, Oregon)

Methods

All layers were projected onto the NAD 1983 HARN State Plane Oregon South coordinate reference system, and all raster layers were calculated with a 10 foot cell size.

Submodel 1: User Demand

The user demand submodel aims to identify locations where e-scooter hubs would get the most use, or where the public needs them most. For this model, I used a residency point layer and workplace point layer from the U.S. Census Bureau (2017), along with a business location point layer from ArcGIS Online (2020), and a bus stop point layer gathered from OpenStreetMap

(2026). I clipped these four layers to the study area, created kernel density estimates for each with a radius of 5000 feet, and normalized each layer. Separately, using a Lane County Census Blocks polygon layer from the U.S. Census Bureau (2020), I calculated population density for each polygon (proportionally adjusting the values for polygons clipped when the layer was clipped to the study area) and converted it to raster, assigning population density to the cell value. I combined these five layers using the raster calculator, using the following equation:

[(residencies * 0.3) + (workplaces * 0.25) + (businesses * 0.25) + (bus stops * 0.2)] * (1 + 0.5 * population density)

This equation serves to assign higher importance to residencies (as nearly all users will have a place of residence), similar weighting to workplaces and businesses (as we anticipate that e-scooter usage will often serve as a commute to work and occasionally for general errands/outings), and a slightly lower weighting to bus stops (as bus stops are fairly prevalent in Eugene and accessible on foot). A "boost" is added to areas with higher population density (without negatively affecting areas with lower population density).

This process produces the following layer, indicating higher user demand in white and lower user demand in black:



Figure 2.1: User Demand Model Output

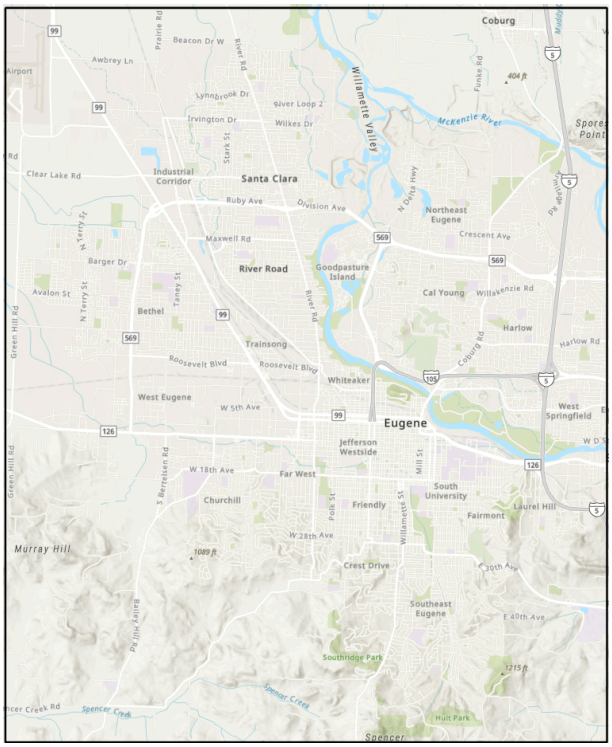


Figure 1: Study Area (for reference)

Submodel 2: Accessibility/Connectivity

The accessibility/connectivity submodel aims to identify locations where e-scooter hubs would be the most accessible and allow users to reach other areas of the study area easily. For this model, I used a bikeways line layer and a crosswalks line layer from the City of Eugene

Mapping Hub (2026, 2025 respectively), as well as the previously mentioned bus stop point layer. I clipped these layers to the study area and created distance accumulation layers for each, where raster cells would indicate increasing distance from these features. I then normalized and inverted the values for these layers using the raster calculator so that cells closer to features would have higher values and cells further away, lower values. I then combined the three layers using the raster calculator with the following equation:

(crosswalks * 0.2) + (bikeways * 0.4) + (bus stop * 0.4)

This equation serves to assign higher importance to bikeways (accessibility and connectivity) and bus stops (connectivity), and a lower importance to crosswalks (connectivity) as bikeway usage is expected to be popular with e-scooters, public transport is highly valued in Eugene, and crosswalks contribute to city/route connectivity (just not quite as much).

This process produces the following layer, indicating higher accessibility/connectivity in white and lower accessibility/connectivity in black:



Figure 2.2: Accessibility Model Output

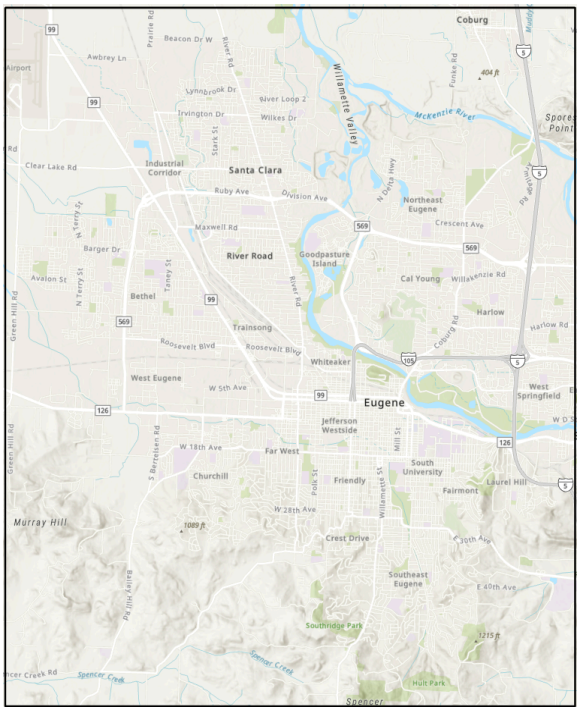


Figure 1: Study Area (for reference)

Submodel 3: Feasibility

The feasibility submodel aims to identify locations where e-scooter hubs can *actually* be placed. For this model, I used the previously mentioned bikeways line layer, along with a sidewalks line layer and a streets line layer from the City of Eugene Mapping Hub (2026). I clipped these layers to the study area and subsetted the streets layer to streets with a speed limit of 30 miles per hour or lower, as hubs may be able to be placed on low-speed streets. I then converted these layers to raster and then used the raster calculator to convert these rasters to

binary, with values of zero where features were not present, and values of one where they were. I then combined the three binary layers to create an output depicting a value of zero (gray) where none of these features were present and a value of one (red) where at least one feature was present:

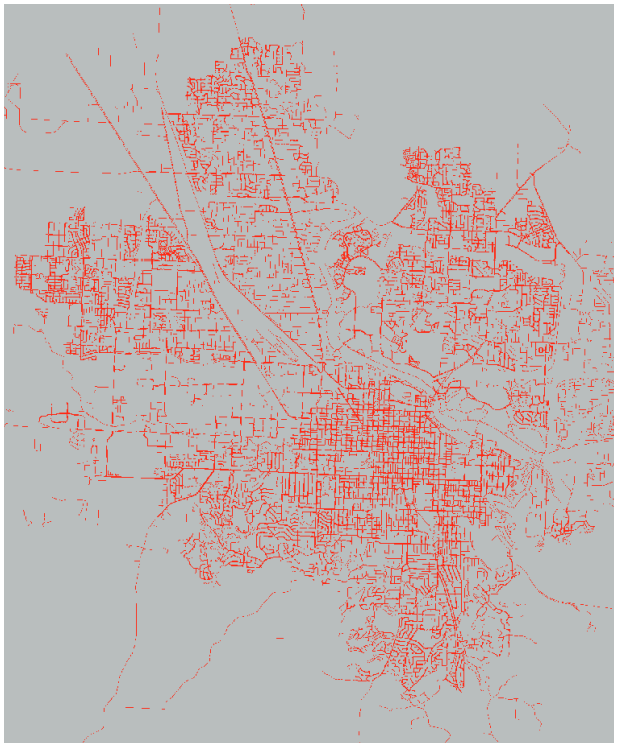


Figure 2.3: Feasibility Model Output

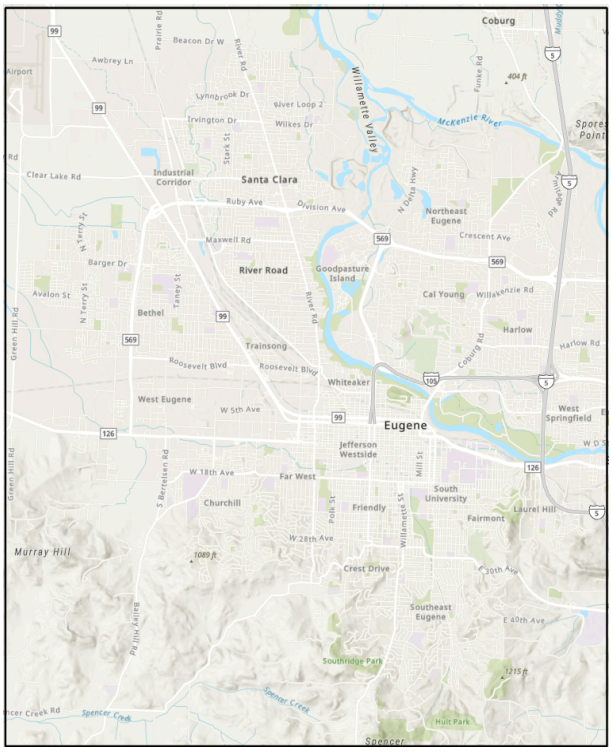


Figure 1: Study Area (for reference)

Combined Model

I started by combining these three outputs using the following equation in the raster calculator:

[(user demand * 0.55) + (accessibility * 0.3)] * feasibility

This allows user demand to be weighted more heavily than accessibility, as I am more interested in finding areas where users are most likely to need the hubs, and still prioritizing (but less so) how optimally-placed the hubs are. Multiplying that by the feasibility binary layer allows us to see that demand only where it is feasible to actually place the hubs. This process resulted in the following layer, indicating higher suitability in white and lower suitability in black:

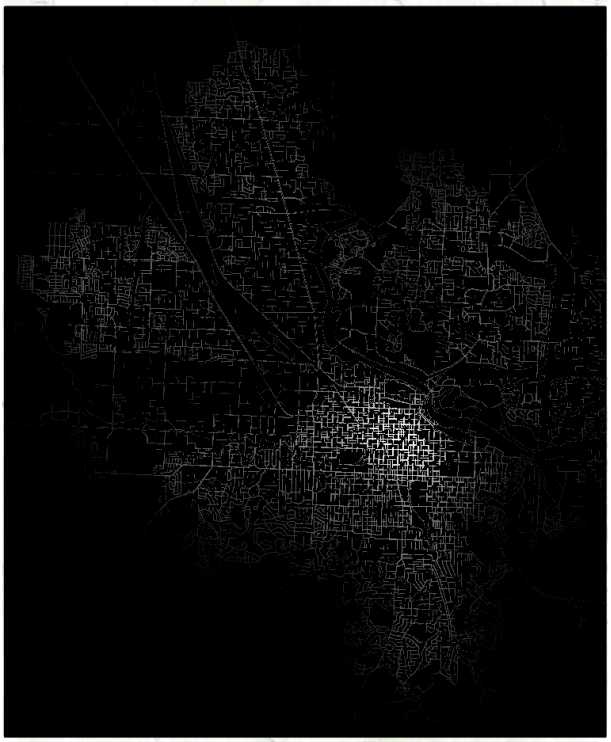


Figure 3.1: Final Model (first attempt)

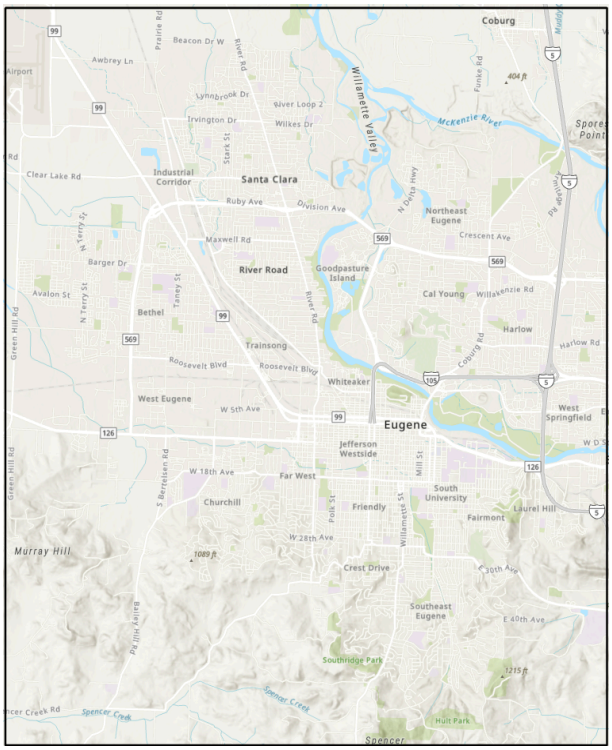


Figure 1: Study Area (for reference)

I was not fond of this output, as the feasibility layer with such a small cell size made it hard to see the continuous color scale when layered on a basemap. As a result, I decided to remove the feasibility layer from the final model and map and instead used the following equation with the raster calculator:

(user demand * 0.55) + (accessibility * 0.3)

This process resulted in the following output indicating higher suitability in white and lower suitability in black:

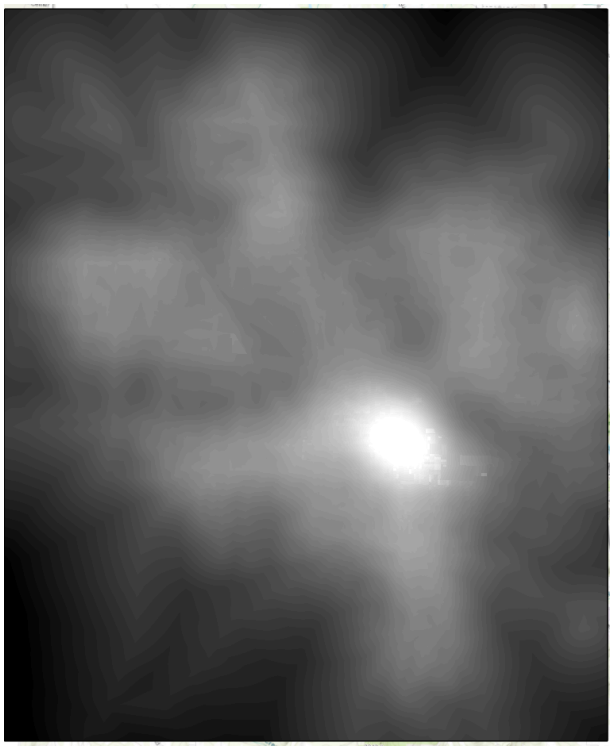


Figure 3.2: Final Model (feasibility removed)

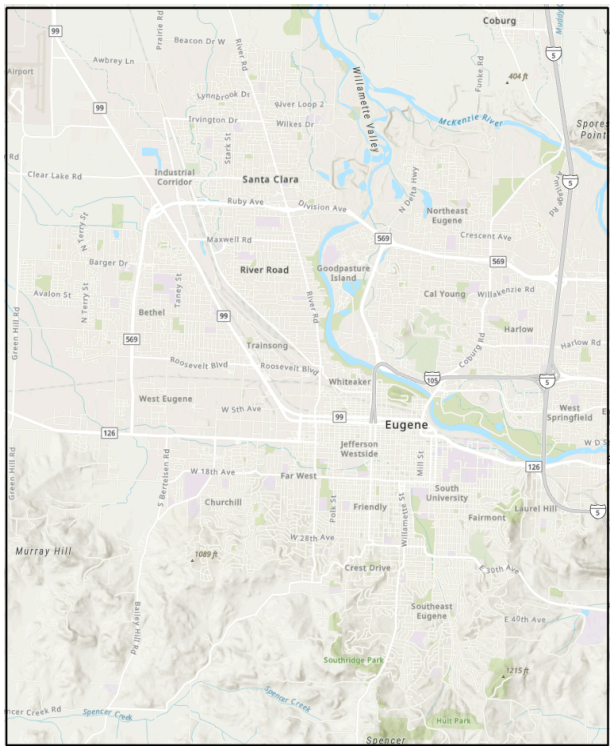


Figure 1: Study Area (for reference)

Using figure 3.2 and applying a classified color mapping with 15 classes (using Jenks Natural Breaks), I created the following map:

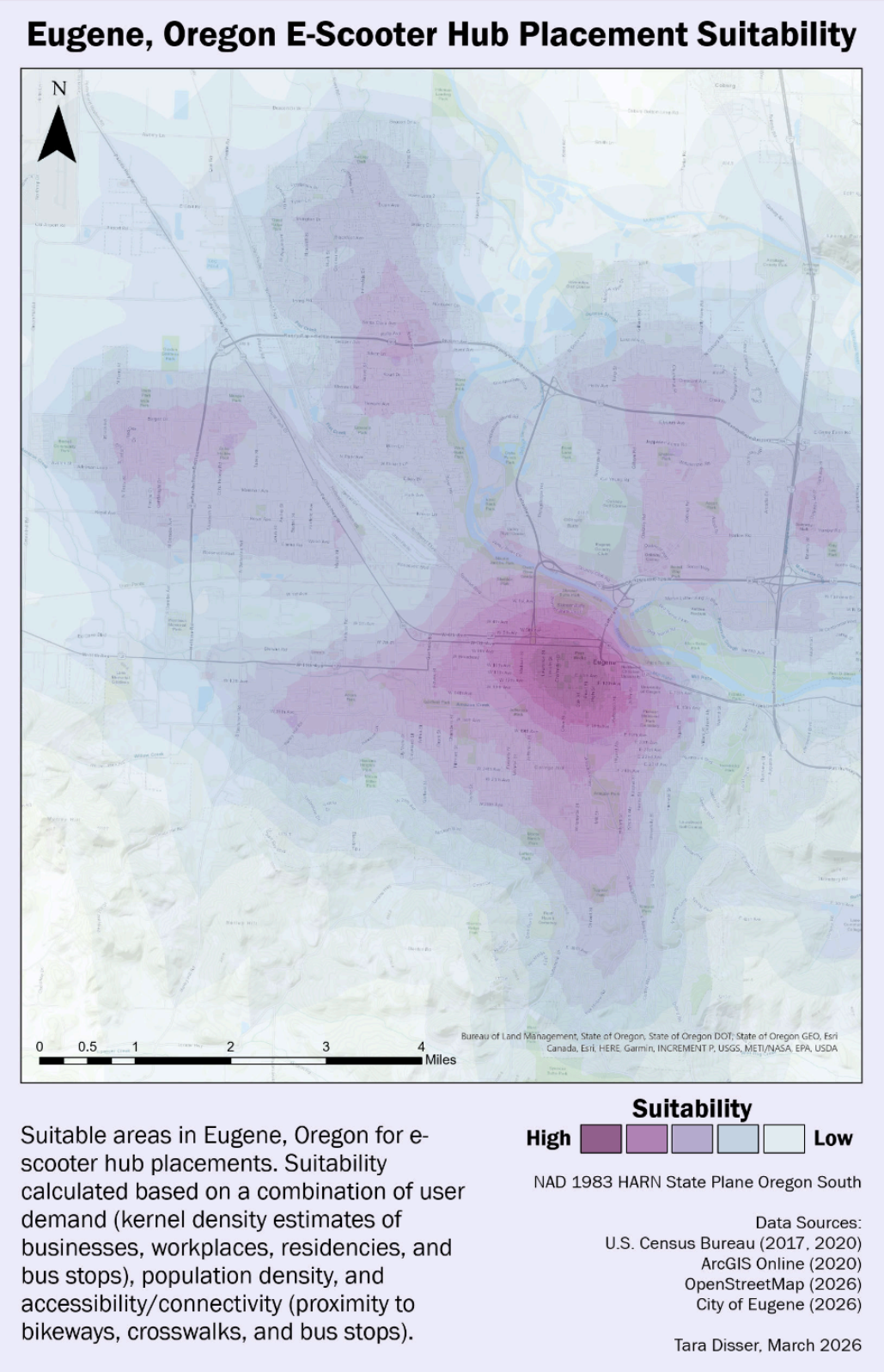


Figure 4: Eugene, Oregon E-Scooter Hub Placement Suitability Map

Assessment of Spatial Patterns

It’s clear to see from this map that downtown Eugene is the most suitable area for e-scooter hub placement. This makes sense, as downtown areas are dense in residencies, businesses, bus stops, population density, sidewalks, streets, and crosswalks. There are simply a lot of people that exist in the downtown area, and thus we find the highest demand there. We also see a lower, but still noticeable level of suitability in a few areas north of downtown as well as directly west and south. These are also areas with higher density of residences, businesses, sidewalks, and streets, simply not quite as much as the downtown area. It would be worthwhile to disperse hubs through these areas with an emphasis on the downtown area.